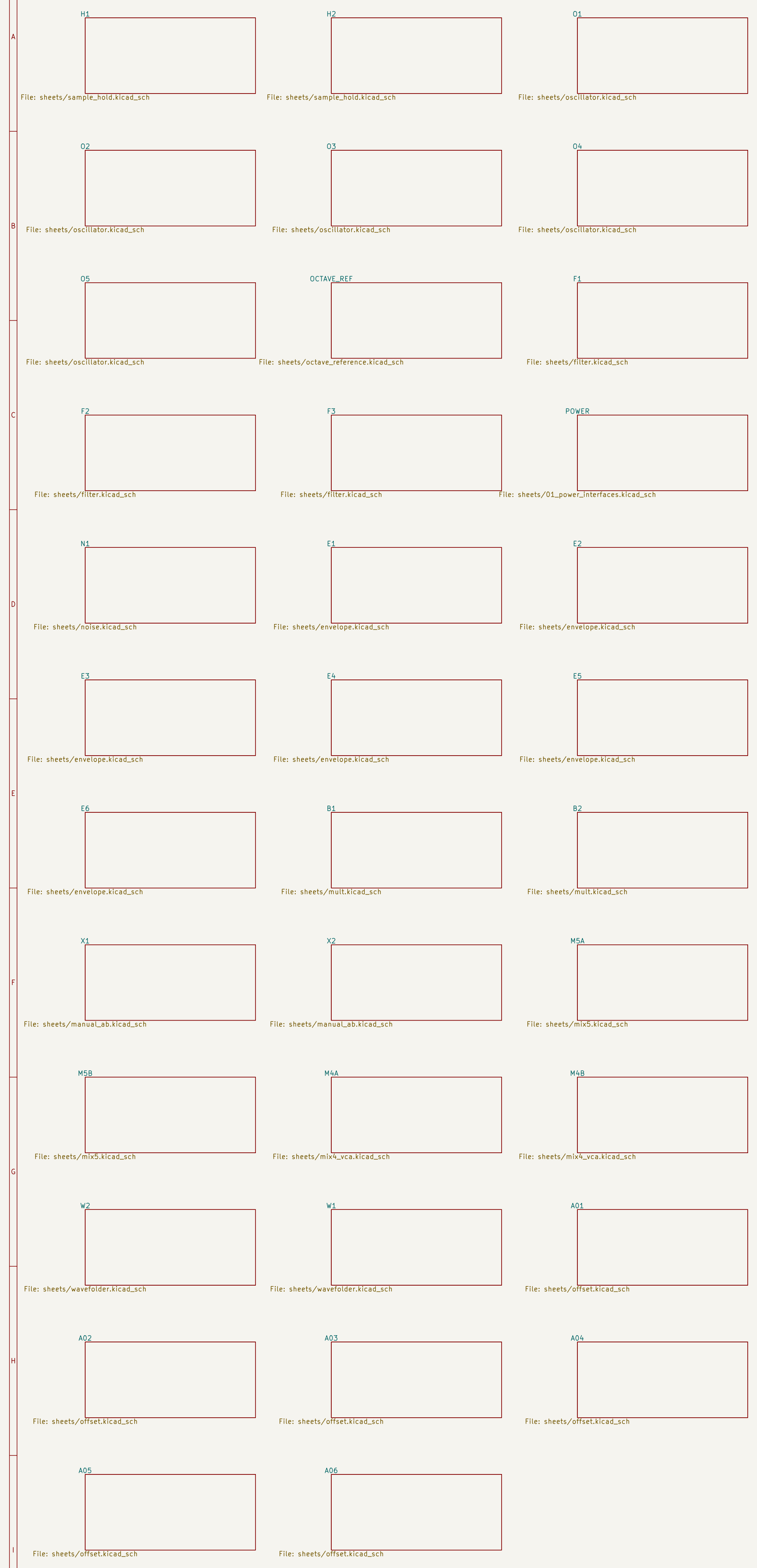
H1
File: sheets/sample_hold.kicad_sch
H2
File: sheets/sample_hold.kicad_sch
O1
File: sheets/oscillator.kicad_sch
O2
File: sheets/oscillator.kicad_sch
O3
File: sheets/oscillator.kicad_sch
O4
File: sheets/oscillator.kicad_sch
O5
File: sheets/oscillator.kicad_sch
OCTAVE_REF
File: sheets/octave_reference.kicad_sch
F1
File: sheets/filter.kicad_sch
F2
File: sheets/filter.kicad_sch
F3
File: sheets/filter.kicad_sch
POWER
File: sheets/01_power_interfaces.kicad_sch
N1
File: sheets/noise.kicad_sch
E1
File: sheets/envelope.kicad_sch
E2
File: sheets/envelope.kicad_sch
E3
File: sheets/envelope.kicad_sch
E4
File: sheets/envelope.kicad_sch
E5
File: sheets/envelope.kicad_sch
E6
File: sheets/envelope.kicad_sch
B1
File: sheets/mult.kicad_sch
B2
File: sheets/mult.kicad_sch
X1
File: sheets/manual_ab.kicad_sch
X2
File: sheets/manual_ab.kicad_sch
M5A
File: sheets/mix5.kicad_sch
M5B
File: sheets/mix5.kicad_sch
M4A
File: sheets/mix4_vca.kicad_sch
M4B
File: sheets/mix4_vca.kicad_sch
W2
File: sheets/wavefolder.kicad_sch
W1
File: sheets/wavefolder.kicad_sch
A01
File: sheets/offset.kicad_sch
A02
File: sheets/offset.kicad_sch
A03
File: sheets/offset.kicad_sch
A04
File: sheets/offset.kicad_sch
A05
File: sheets/offset.kicad_sch
A06
File: sheets/offset.kicad_sch

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Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
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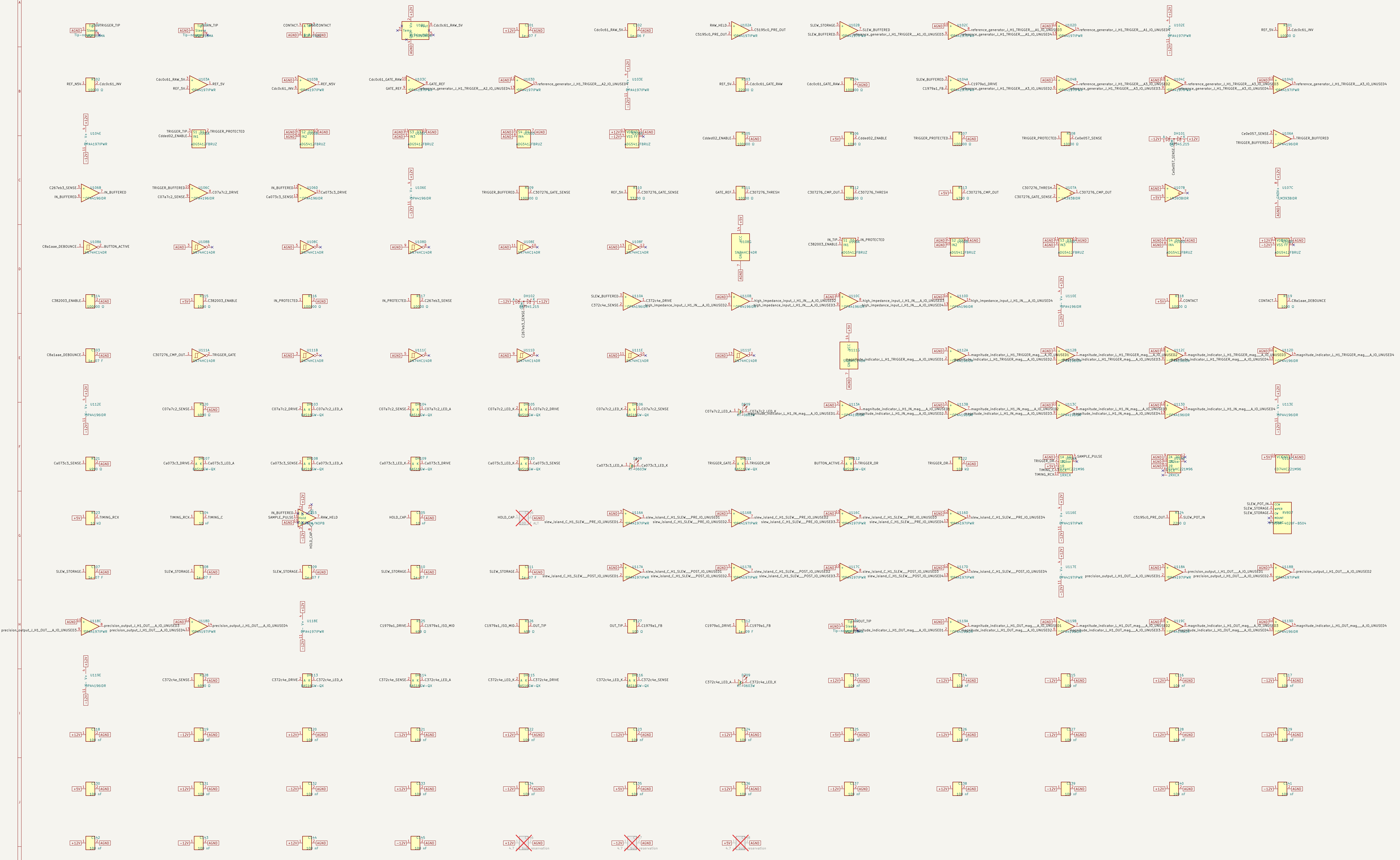

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RAW_HELD
C5195c0_PRE_OUT
SLEW_STORAGE
SLEW_BUFFERED
REF_5V
Cdc0c61_INV
REF_N5V
Cdc0c61_RAW_5V
Cdc0c61_GATE_RAW
GATE_REF
TRIGGER_PROTECTED
Cdded02_ENABLE
Ce0e057_SENSE
TRIGGER_BUFFERED
C267eb3_SENSE
IN_BUFFERED
C07a7c2_DRIVE
Ca073c3_DRIVE
C307276_GATE_SENSE
C307276_THRESH
C307276_CMP_OUT
C8a1aae_DEBOUNCE
BUTTON_ACTIVE
IN_PROTECTED
C382003_ENABLE
CONTACT
TRIGGER_GATE
SAMPLE_PULSE
TIMING_RCX
TIMING_C
HOLD_CAP
SLEW_POT_IN
OUT_TIP
C1979a1_DRIVE
C1979a1_ISO_MID
C1979a1_FB
C372c4e_DRIVE
C372c4e_SENSE
OPA4197IPWR
OPA4196IDR
SN74HC14DR
ADG5412FBRUZ
LM393BIDR
SN74HC221M96
BAS16W-QX
BAT54S,215
AGND
+12V
-12V
+5V

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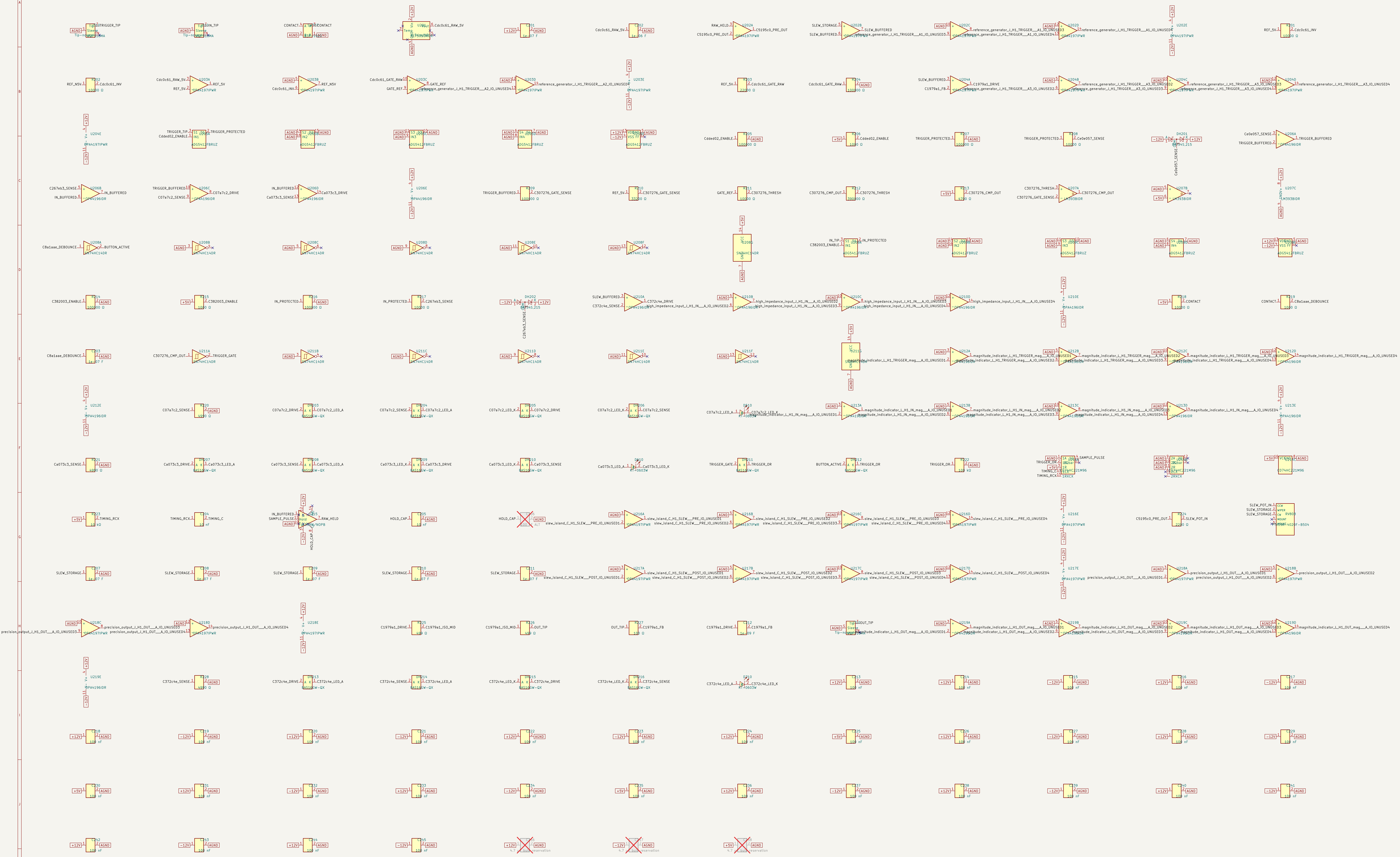
Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.6
Id: 2/36



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KiCad E.D.A. 10.0.6		Id: 3/36

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Size: A0 | Date: | Rev:

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Title:

Size: A0 Date: **Rev:**

KiCad E.D.A. 10.0.6 Id: 5/36

Sheet: /03/
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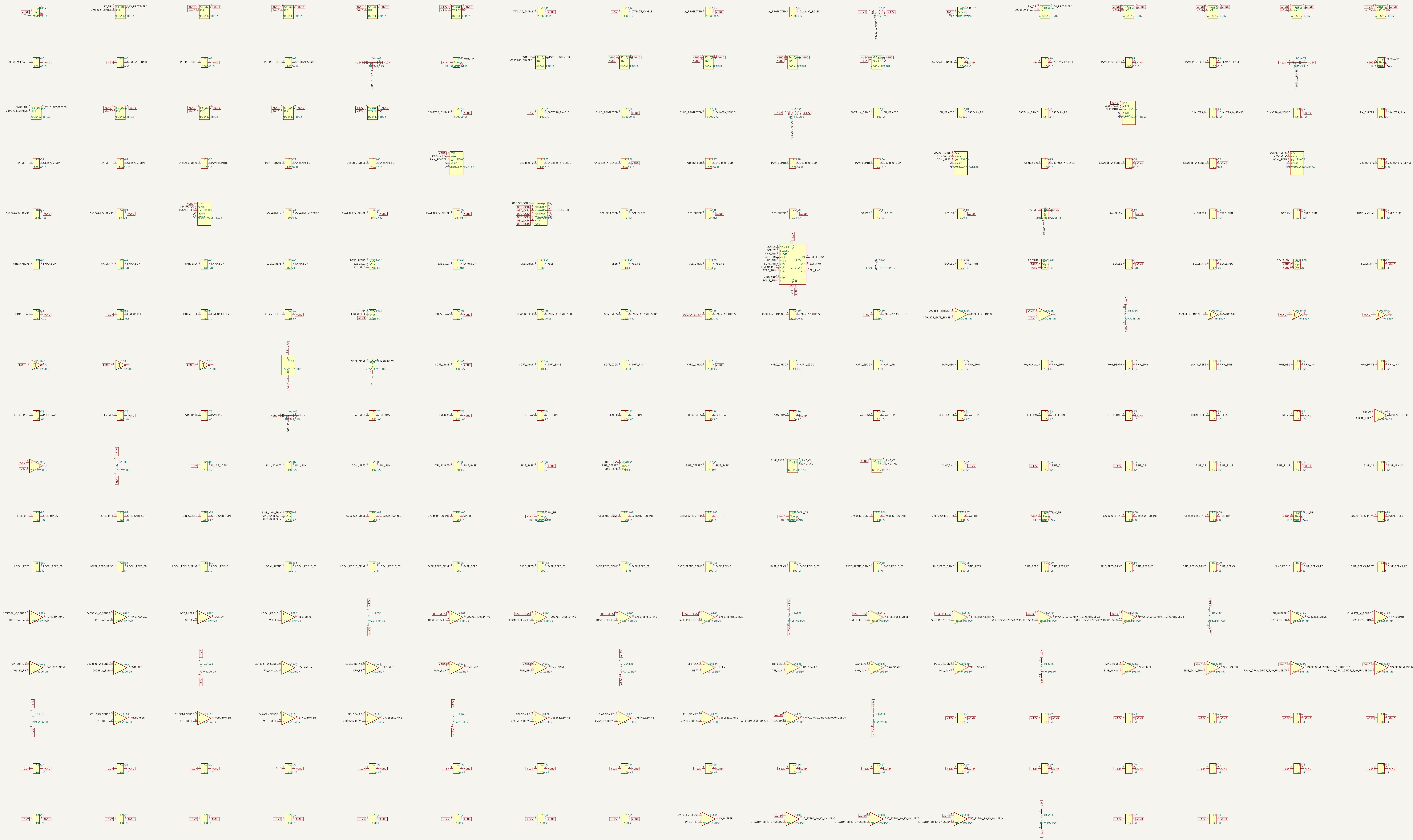
KiCad E.D.A. 10.0.6



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KiCad E.D.A. 10.0.6		Id: 7/36

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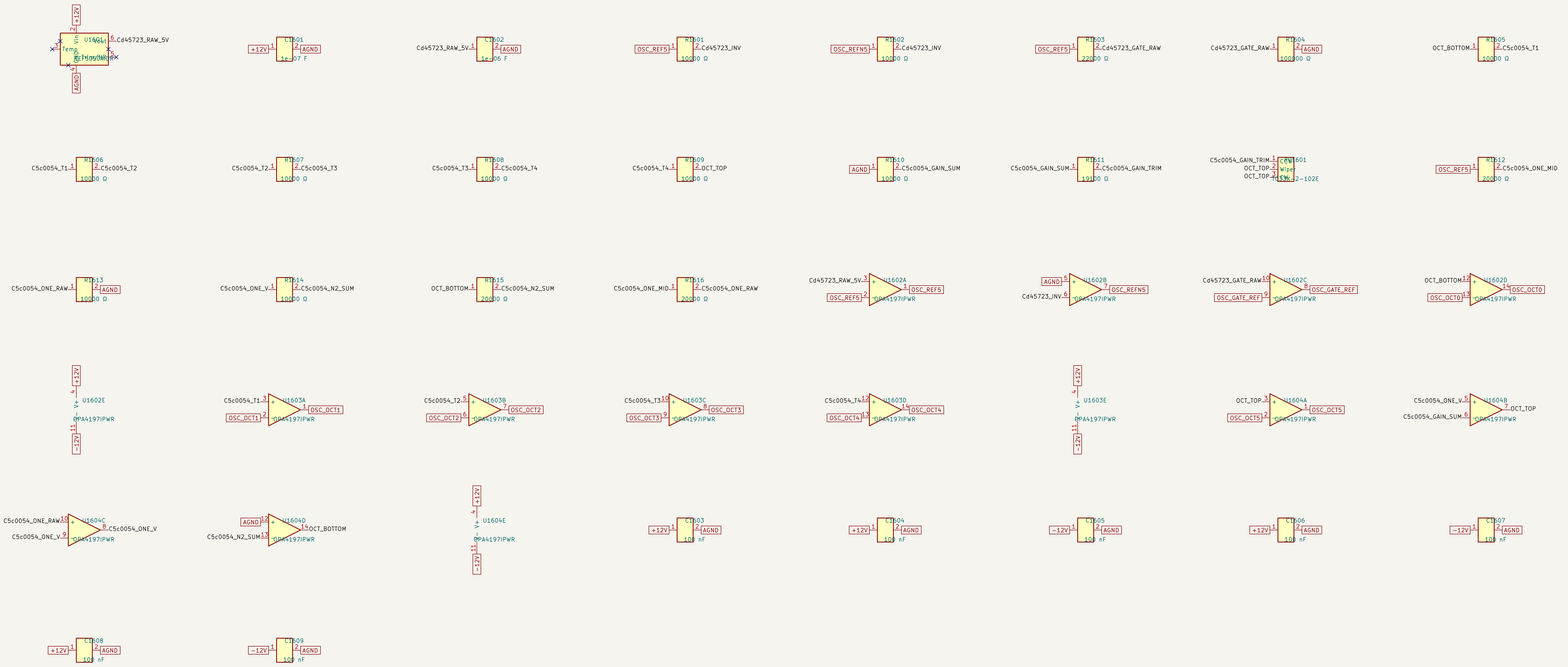
Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 8/36



+12V
U1601
Vin
Vout
Temp
GND
Trim/NR
REF5050AIDR
AGND
Cd45723_RAW_5V
C1601
+12V
AGND
1e-07 F
C1602
Cd45723_RAW_5V
AGND
1e-06 F
R1601
OSC_REF5
Cd45723_INV
10000 Ω
R1602
OSC_REFN5
Cd45723_INV
10000 Ω
R1603
OSC_REF5
Cd45723_GATE_RAW
22000 Ω
R1604
Cd45723_GATE_RAW
AGND
100000 Ω
R1605
OCT_BOTTOM
C5c0054_T1
10000 Ω
R1606
C5c0054_T1
C5c0054_T2
10000 Ω
R1607
C5c0054_T2
C5c0054_T3
10000 Ω
R1608
C5c0054_T3
C5c0054_T4
10000 Ω
R1609
C5c0054_T4
OCT_TOP
10000 Ω
R1610
AGND
C5c0054_GAIN_SUM
10000 Ω
R1611
C5c0054_GAIN_SUM
C5c0054_GAIN_TRIM
19100 Ω
RV1601
C5c0054_GAIN_TRIM
OCT_TOP
OCT_TOP
CCW
Wiper
CW
TC33X-2-102E
R1612
OSC_REF5
C5c0054_ONE_MID
20000 Ω
R1613
C5c0054_ONE_RAW
AGND
10000 Ω
R1614
C5c0054_ONE_V
C5c0054_N2_SUM
10000 Ω
R1615
OCT_BOTTOM
C5c0054_N2_SUM
20000 Ω
R1616
C5c0054_ONE_MID
C5c0054_ONE_RAW
20000 Ω
U1602A
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OSC_REF5
OPA4197IPWR
U1602B
AGND
Cd45723_INV
OSC_REFN5
OPA4197IPWR
U1602C
Cd45723_GATE_RAW
OSC_GATE_REF
OSC_GATE_REF
OPA4197IPWR
U1602D
OCT_BOTTOM
OSC_OCT0
OSC_OCT0
OPA4197IPWR
+12V
U1602E
V+
V-
OPA4197IPWR
-12V
U1603A
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OSC_OCT1
OSC_OCT1
OPA4197IPWR
U1603B
C5c0054_T2
OSC_OCT2
OSC_OCT2
OPA4197IPWR
U1603C
C5c0054_T3
OSC_OCT3
OSC_OCT3
OPA4197IPWR
U1603D
C5c0054_T4
OSC_OCT4
OSC_OCT4
OPA4197IPWR
+12V
U1603E
V+
V-
OPA4197IPWR
-12V
U1604A
OCT_TOP
OSC_OCT5
OSC_OCT5
OPA4197IPWR
U1604B
C5c0054_ONE_V
C5c0054_GAIN_SUM
OCT_TOP
OPA4197IPWR
U1604C
C5c0054_ONE_RAW
C5c0054_ONE_V
C5c0054_ONE_V
OPA4197IPWR
U1604D
AGND
C5c0054_N2_SUM
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OPA4197IPWR
+12V
U1604E
V+
V-
OPA4197IPWR
-12V
C1603
+12V
AGND
100 nF
C1604
+12V
AGND
100 nF
C1605
-12V
AGND
100 nF
C1606
+12V
AGND
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-12V
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AGND
100 nF
C1609
-12V
AGND
100 nF

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Rev:
KiCad E.D.A. 10.0.6
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Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 10/36

Sheet: /F2/
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Size: A0 Date: **Rev:**
KiCad E.D.A. 10.0.6 Id: 11/36

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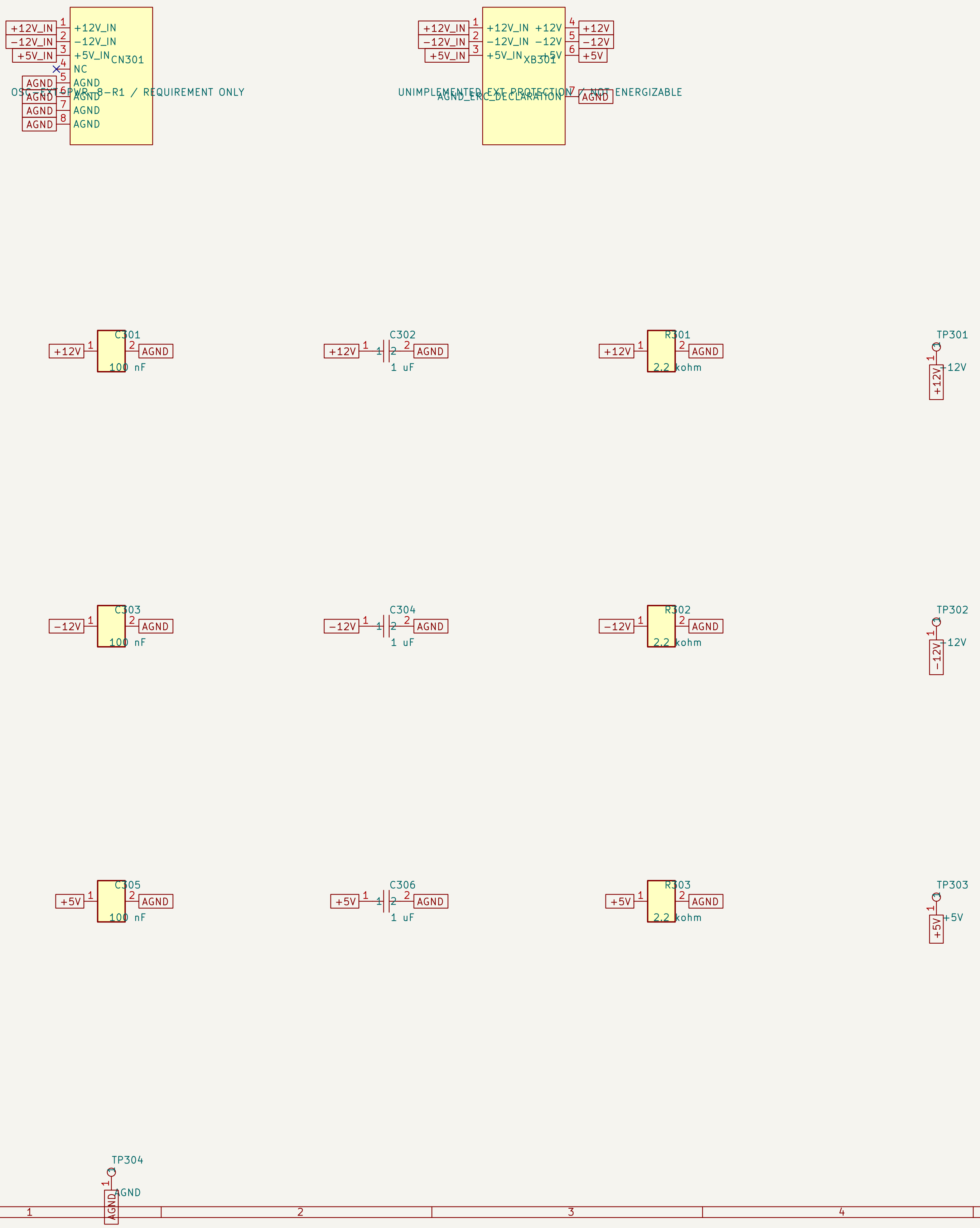
Rev:
Id: 12/36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+5V_IN
AGND
AGND
AGND
AGND
1
2
3
4
5
6
7
8
+12V_IN
−12V_IN
+5V_IN
NC
AGND
AGND
AGND
AGND
CN301
OSC−EXT−PWR−8−R1 / REQUIREMENT ONLY
+12V_IN
−12V_IN
+5V_IN
1
2
3
+12V_IN
−12V_IN
+5V_IN
+12V
−12V
+5V
4
5
6
+12V
−12V
+5V
XB301
UNIMPLEMENTED EXT PROTECTION / NOT ENERGIZABLE
AGND_ERC_DECLARATION
7
AGND
C301
+12V
1
2
AGND
100 nF
C302
+12V
1
2
AGND
1 uF
R301
+12V
1
2
AGND
2.2 kohm
TP301
1
+12V
+12V
C303
−12V
1
2
AGND
100 nF
C304
−12V
1
2
AGND
1 uF
R302
−12V
1
2
AGND
2.2 kohm
TP302
1
−12V
−12V
C305
+5V
1
2
AGND
100 nF
C306
+5V
1
2
AGND
1 uF
R303
+5V
1
2
AGND
2.2 kohm
TP303
1
+5V
+5V
TP304
1
AGND
AGND

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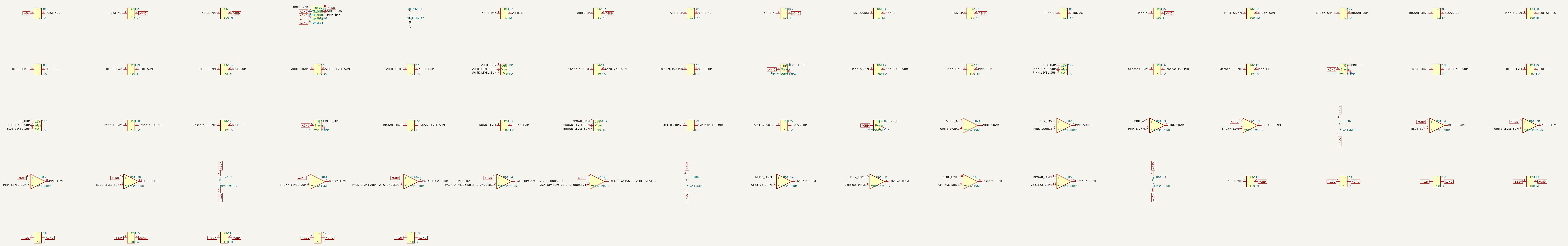
Title:

Size: A3 Date:

Rev:

KiCad E.D.A. 10.0.6

Id: 13/36



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C6101 NOISE_VDD AGND 4.7 µF
C6102 NOISE_VDD AGND 100 nF
U6101 NOISE2 NOISE_VDD +5V 0V AGND WHITE_OUT PINK_OUT WHITE_RAW PINK_RAW Unused
#FLG6101 NOISE_VDD FILTERED_5V
R6102 WHITE_RAW WHITE_LP 1 kΩ
C6103 WHITE_LP AGND 10 nF
C6104 WHITE_LP WHITE_AC 100 nF
R6103 WHITE_AC AGND 100 kΩ
R6104 PINK_SOURCE PINK_LP 1 kΩ
C6105 PINK_LP AGND 10 nF
C6106 PINK_LP PINK_AC 100 nF
R6105 PINK_AC AGND 100 kΩ
R6106 WHITE_SIGNAL BROWN_SUM 100 kΩ
R6107 BROWN_SHAPE BROWN_SUM 1 MΩ
C6107 BROWN_SHAPE BROWN_SUM 10 nF
C6108 PINK_SIGNAL BLUE_SERIES 100 pF
R6108 BLUE_SERIES BLUE_SUM 100 kΩ
R6109 BLUE_SHAPE BLUE_SUM 100 kΩ
C6109 BLUE_SHAPE BLUE_SUM 33 pF
R6110 WHITE_SIGNAL WHITE_LEVEL_SUM 100 kΩ
R6111 WHITE_LEVEL WHITE_TRIM 100 kΩ
RV6101 WHITE_TRIM WHITE_LEVEL_SUM CCW Wiper CW 10 kΩ
R6112 Cbe877b_DRIVE Cbe877b_ISO_MID 499 Ω
R6113 Cbe877b_ISO_MID WHITE_TIP 499 Ω
J1109 AGND S T TN Tip Sleeve Tip-normal WQP518MA WHITE_TIP
R6114 PINK_SIGNAL PINK_LEVEL_SUM 100 kΩ
R6115 PINK_LEVEL PINK_TRIM 100 kΩ
RV6102 PINK_TRIM PINK_LEVEL_SUM CCW Wiper CW 10 kΩ
R6116 Cdbc5aa_DRIVE Cdbc5aa_ISO_MID 499 Ω
R6117 Cdbc5aa_ISO_MID PINK_TIP 499 Ω
AGND Tip Sleeve Tip-normal WQP518MA PINK_TIP
R6118 BLUE_SHAPE BLUE_LEVEL_SUM 10 kΩ
R6119 BLUE_LEVEL BLUE_TRIM 100 kΩ
RV6103 BLUE_TRIM BLUE_LEVEL_SUM CCW Wiper CW 10 kΩ
R6120 Ce44f9a_DRIVE Ce44f9a_ISO_MID 499 Ω
R6121 Ce44f9a_ISO_MID BLUE_TIP 499 Ω
AGND Tip Sleeve Tip-normal WQP518MA BLUE_TIP
R6122 BROWN_SHAPE BROWN_LEVEL_SUM 20 kΩ
R6123 BROWN_LEVEL BROWN_TRIM 100 kΩ
RV6104 BROWN_TRIM BROWN_LEVEL_SUM CCW Wiper CW 10 kΩ
R6124 Cde1183_DRIVE Cde1183_ISO_MID 499 Ω
R6125 Cde1183_ISO_MID BROWN_TIP 499 Ω
AGND Tip Sleeve Tip-normal WQP518MA BROWN_TIP
U6102A OPA4196IDR WHITE_AC WHITE_SIGNAL
U6102B OPA4196IDR PINK_RAW PINK_SOURCE
U6102C OPA4196IDR PINK_AC PINK_SIGNAL
U6102D OPA4196IDR AGND BROWN_SUM BROWN_SHAPE
U6102E OPA4196IDR +12V −12V V+ V−
U6103A OPA4196IDR AGND BLUE_SUM BLUE_SHAPE
U6103B OPA4196IDR AGND WHITE_LEVEL_SUM WHITE_LEVEL
U6103C OPA4196IDR AGND PINK_LEVEL_SUM PINK_LEVEL
U6103D OPA4196IDR AGND BLUE_LEVEL_SUM BLUE_LEVEL
U6103E OPA4196IDR +12V −12V V+ V−
U6104A OPA4196IDR AGND BROWN_LEVEL_SUM BROWN_LEVEL
U6104B OPA4196IDR AGND PACK_OPA4196IDR_2_IO_UNUSED2
U6104C OPA4196IDR AGND PACK_OPA4196IDR_2_IO_UNUSED3
U6104D OPA4196IDR AGND PACK_OPA4196IDR_2_IO_UNUSED4
U6104E OPA4196IDR +12V −12V V+ V−
U6105A OPA4196IDR WHITE_LEVEL Cbe877b_DRIVE
U6105B OPA4196IDR PINK_LEVEL Cdbc5aa_DRIVE
U6105C OPA4196IDR BLUE_LEVEL Ce44f9a_DRIVE
U6105D OPA4196IDR BROWN_LEVEL Cde1183_DRIVE
U6105E OPA4196IDR +12V −12V V+ V−
C6110 NOISE_VDD AGND 100 nF
C6111 +12V AGND 100 nF
C6112 −12V AGND 100 nF
C6113 +12V AGND 100 nF
C6114 −12V AGND 100 nF
C6115 +12V AGND 100 nF
C6116 −12V AGND 100 nF
C6117 +12V AGND 100 nF
C6118 −12V AGND 100 nF

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Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
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Sheet: /E1/
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Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 15/36

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KiCad E.D.A. 10.0.6 Id: 16/36

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Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 17/36

Sheet: /E4/
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Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 18/36

Sheet: /E5/
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Title:

Size: A0 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 19/36

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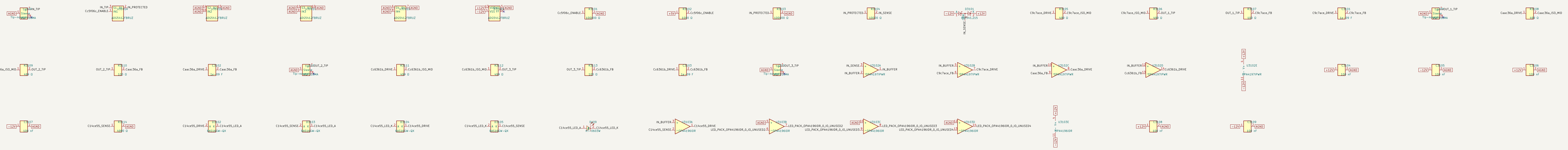
KiCad E.D.A. 10.0.6 | Id: 20/36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Tip
Sleeve
Tip−normal
WQP518MA
IN_TIP
J109
IN_TIP
Cc5f06c_ENABLE
U3101A
ADG5412FBRUZ
IN_PROTECTED
U3101B
U3101C
U3101D
U3101E
+12V
−12V
VDD
GND
VSS
FF
R3101
100000 Ω
R3102
1000 Ω
+5V
R3103
R3104
10000 Ω
IN_SENSE
D3101
BAT54S,215
R3105
499 Ω
C9c7ace_DRIVE
C9c7ace_ISO_MID
R3106
OUT_1_TIP
R3107
100 Ω
C9c7ace_FB
C3101
1e−09 F
R3108
Caac36a_DRIVE
Caac36a_ISO_MID
R3109
OUT_2_TIP
R3110
Caac36a_FB
C3102
J110
R3111
Cc6361b_DRIVE
Cc6361b_ISO_MID
R3112
OUT_3_TIP
R3113
Cc6361b_FB
C3103
U3102A
OPA4197IPWR
IN_BUFFER
U3102B
U3102C
U3102D
U3102E
C3104
100 nF
C3105
C3106
C3107
R3114
4990 Ω
C14ce55_SENSE
D3102
BAS16GW−QX
C14ce55_DRIVE
C14ce55_LED_A
D3103
D3104
C14ce55_LED_K
D3105
D109
KT−0603W
U3103A
OPA4196IDR
U3103B
U3103C
U3103D
U3103E
LED_PACK_OPA4196IDR_0_IO_UNUSED2
LED_PACK_OPA4196IDR_0_IO_UNUSED3
LED_PACK_OPA4196IDR_0_IO_UNUSED4
C3108
C3109

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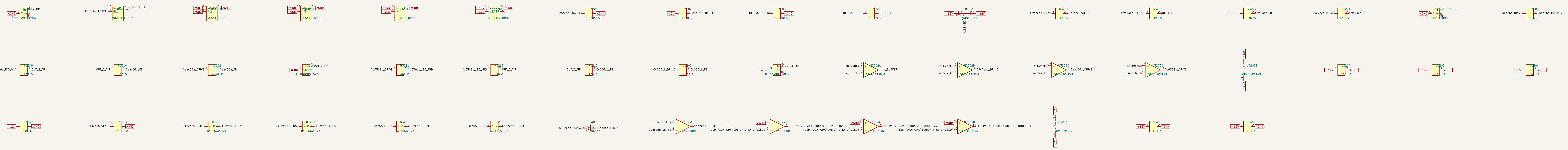
Size: A0 | Date: | **Rev:**

KiCad E.D.A. 10.0.6 | Id: 21/36


AGND
Tip
Sleeve
TN
IN_TIP
Tip–normal
U3201A
IN_TIP
S1
D1
IN1
IN_PROTECTED
Cc5f06c_ENABLE
ADG5412FBRUZ
U3201B
AGND
S2
D2
IN2
ADG5412FBRUZ
U3201C
S3
D3
IN3
ADG5412FBRUZ
U3201D
S4
D4
IN4
ADG5412FBRUZ
U3201E
+12V
−12V
VDD
GND
VSS
ADG5412FBRUZ
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Cc5f06c_ENABLE
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R3203
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R3204
IN_SENSE
10000 Ω
D3201
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K
BAT54S,215
R3205
C9c7ace_DRIVE
C9c7ace_ISO_MID
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R3206
OUT_1_TIP
499 Ω
R3207
C9c7ace_FB
100 Ω
C3201
1e−09 F
R3208
Caac36a_DRIVE
Caac36a_ISO_MID
499 Ω
R3209
OUT_2_TIP
499 Ω
R3210
Caac36a_FB
100 Ω
C3202
1e−09 F
R3211
Cc6361b_DRIVE
Cc6361b_ISO_MID
499 Ω
R3212
OUT_3_TIP
499 Ω
R3213
Cc6361b_FB
100 Ω
C3203
1e−09 F
U3202A
IN_SENSE
IN_BUFFER
OPA4197IPWR
U3202B
OPA4197IPWR
U3202C
OPA4197IPWR
U3202D
OPA4197IPWR
U3202E
V+
V−
OPA4197IPWR
C3204
100 nF
C3205
100 nF
C3206
100 nF
C3207
100 nF
R3214
C14ce55_SENSE
4990 Ω
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C14ce55_LED_A
BAS16GW−QX
D3203
BAS16GW−QX
D3204
C14ce55_LED_K
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D3205
BAS16GW−QX
KT−0603W
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U3203B
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OPA4196IDR
U3203C
LED_PACK_OPA4196IDR_0_IO_UNUSED3
OPA4196IDR
U3203D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
OPA4196IDR
U3203E
OPA4196IDR
C3208
100 nF
C3209
100 nF

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File: mult.kicad_sch

Title:

Size: A0
Date:
Rev:
KiCad E.D.A. 10.0.6
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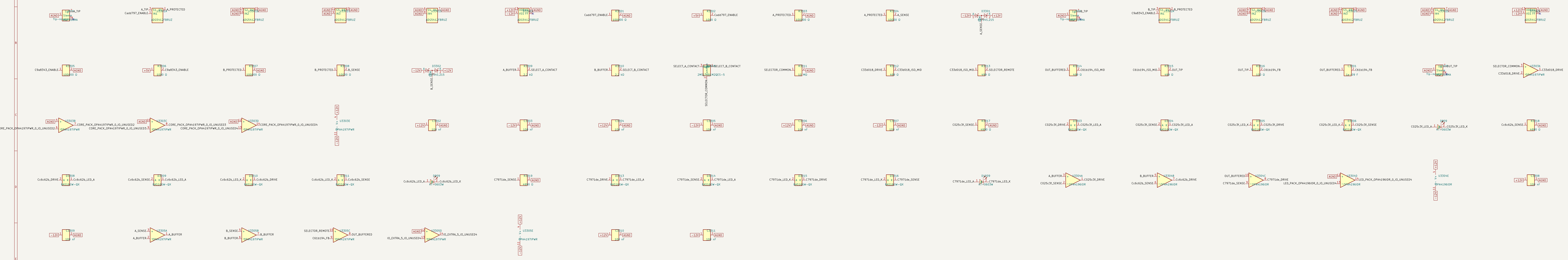
J3301 A_TIP Tip Sleeve AGND
U3301A A_TIP A_PROTECTED Cadd797_ENABLE ADG5412FBRUZ
U3301B AGND ADG5412FBRUZ
U3301C AGND ADG5412FBRUZ
U3301D AGND ADG5412FBRUZ
U3301E +12V -12V VDD GND VSS AGND ADG5412FBRUZ
R3301 Cadd797_ENABLE AGND 100000 Ω
R3302 +5V Cadd797_ENABLE 1000 Ω
R3303 A_PROTECTED AGND 100000 Ω
R3304 A_PROTECTED A_SENSE 10000 Ω
D3301 -12V +12V A_SENSE BAT54S,215
J3302 B_TIP Tip Sleeve AGND
U3302A B_TIP B_PROTECTED C9a8343_ENABLE ADG5412FBRUZ
U3302B AGND ADG5412FBRUZ
U3302C AGND ADG5412FBRUZ
U3302D AGND ADG5412FBRUZ
U3302E +12V -12V AGND ADG5412FBRUZ
R3305 C9a8343_ENABLE AGND 100000 Ω
R3306 +5V C9a8343_ENABLE 1000 Ω
R3307 B_PROTECTED AGND 100000 Ω
R3308 B_PROTECTED B_SENSE 10000 Ω
D3302 -12V +12V B_SENSE BAT54S,215
R3309 A_BUFFER SELECT_A_CONTACT 2.2 kΩ
R3310 B_BUFFER SELECT_B_CONTACT 2.2 kΩ
SW3301 SELECT_A_CONTACT SELECT_B_CONTACT SELECTOR_COMMON 2MS1T1B1M2QES-5
R3311 SELECTOR_COMMON AGND 10 MΩ
R3312 C33d018_DRIVE C33d018_ISO_MID 499 Ω
R3313 C33d018_ISO_MID SELECTOR_REMOTE 499 Ω
R3314 OUT_BUFFERED C61b194_ISO_MID 499 Ω
R3315 C61b194_ISO_MID OUT_TIP 499 Ω
R3316 OUT_TIP C61b194_FB 100 Ω
C3301 OUT_BUFFERED C61b194_FB 1e-09 F
J3303 OUT_TIP Tip Sleeve AGND
U3303A SELECTOR_COMMON C33d018_DRIVE OPA4197IPWR
U3303B AGND CORE_PACK_OPA4197IPWR_0_IO_UNUSED2 OPA4197IPWR
U3303C AGND CORE_PACK_OPA4197IPWR_0_IO_UNUSED3 OPA4197IPWR
U3303D AGND CORE_PACK_OPA4197IPWR_0_IO_UNUSED4 OPA4197IPWR
U3303E +12V -12V OPA4197IPWR
C3302 +12V AGND 100 nF
C3303 -12V AGND 100 nF
C3304 +12V AGND 100 nF
C3305 -12V AGND 100 nF
C3306 +12V AGND 100 nF
C3307 -12V AGND 100 nF
R3317 C025c3f_SENSE AGND 4990 Ω
D3303 C025c3f_DRIVE C025c3f_LED_A BAS16GW-QX
D3304 C025c3f_SENSE C025c3f_LED_A BAS16GW-QX
D3305 C025c3f_LED_K C025c3f_DRIVE BAS16GW-QX
D3306 C025c3f_LED_K C025c3f_SENSE BAS16GW-QX
D909 C025c3f_LED_A C025c3f_LED_K KT-0603W
R3318 Cc6c62b_SENSE AGND 4990 Ω
D3308 Cc6c62b_DRIVE Cc6c62b_LED_A BAS16GW-QX
D3309 Cc6c62b_SENSE Cc6c62b_LED_A BAS16GW-QX
D3310 Cc6c62b_LED_K Cc6c62b_DRIVE BAS16GW-QX
D3311 Cc6c62b_LED_K Cc6c62b_SENSE BAS16GW-QX
D909 Cc6c62b_LED_A Cc6c62b_LED_K KT-0603W
R3319 C7971de_SENSE AGND 4990 Ω
D3313 C7971de_DRIVE C7971de_LED_A BAS16GW-QX
D3314 C7971de_SENSE C7971de_LED_A BAS16GW-QX
D3315 C7971de_LED_K C7971de_DRIVE BAS16GW-QX
D3316 C7971de_LED_K C7971de_SENSE BAS16GW-QX
D1009 C7971de_LED_A C7971de_LED_K KT-0603W
U3304A A_BUFFER C025c3f_SENSE C025c3f_DRIVE OPA4196IDR
U3304B B_BUFFER Cc6c62b_SENSE Cc6c62b_DRIVE OPA4196IDR
U3304C OUT_BUFFERED C7971de_SENSE C7971de_DRIVE OPA4196IDR
U3304D AGND LED_PACK_OPA4196IDR_0_IO_UNUSED4 OPA4196IDR
U3304E +12V -12V OPA4196IDR
C3308 +12V AGND 100 nF
C3309 -12V AGND 100 nF
U3305A A_SENSE A_BUFFER OPA4197IPWR
U3305B B_SENSE B_BUFFER OPA4197IPWR
U3305C SELECTOR_REMOTE C61b194_FB OUT_BUFFERED OPA4197IPWR
U3305D AGND IO_EXTRA_5_IO_UNUSED4 OPA4197IPWR
U3305E +12V -12V OPA4197IPWR
C3310 +12V AGND 100 nF
C3311 -12V AGND 100 nF

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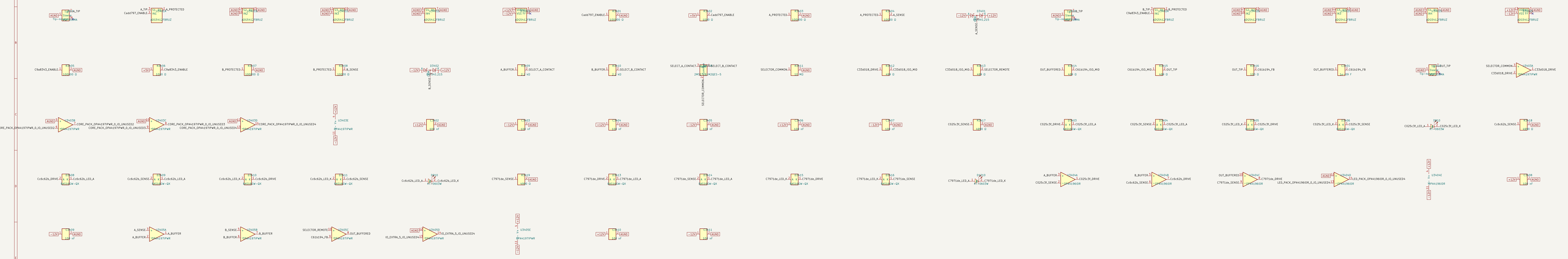
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AGND
Tip
Sleeve
Tip-normal
TN
A_TIP
Cadd797_ENABLE
S1 D1
IN1
U3401A
A_PROTECTED
ADG5412FBRUZ
AGND
S2 D2
IN2
U3401B
ADG5412FBRUZ
S3 D3
IN3
U3401C
ADG5412FBRUZ
S4 D4
IN4
U3401D
ADG5412FBRUZ
+12V
-12V
VDD GND
VSS FF
U3401E
ADG5412FBRUZ
R3401
100000 Ω
R3402
+5V
1000 Ω
R3403
100000 Ω
R3404
A_SENSE
10000 Ω
D3401
BAT54S,215
B_TIP
B_TIP
C9a8343_ENABLE
U3402A
B_PROTECTED
U3402B
U3402C
U3402D
U3402E
R3405
C9a8343_ENABLE
100000 Ω
R3406
1000 Ω
R3407
B_PROTECTED
100000 Ω
R3408
B_SENSE
10000 Ω
D3402
BAT54S,215
B_SENSE
R3409
A_BUFFER
SELECT_A_CONTACT
2.2 kΩ
R3410
B_BUFFER
SELECT_B_CONTACT
2.2 kΩ
SELECT_A_CONTACT
SELECT_B_CONTACT
2MS1T1B1M2QES-5
SELECTOR_COMMON
R3411
SELECTOR_COMMON
10 MΩ
R3412
C33d018_DRIVE
C33d018_ISO_MID
499 Ω
R3413
C33d018_ISO_MID
SELECTOR_REMOTE
499 Ω
R3414
OUT_BUFFERED
C61b194_ISO_MID
499 Ω
R3415
C61b194_ISO_MID
OUT_TIP
499 Ω
R3416
OUT_TIP
C61b194_FB
100 Ω
C3401
OUT_BUFFERED
C61b194_FB
1e-09 F
OUT_TIP
SELECTOR_COMMON
C33d018_DRIVE
U3403A
C33d018_DRIVE
OPA4197IPWR
U3403B
CORE_PACK_OPA4197IPWR_0_IO_UNUSED2
CORE_PACK_OPA4197IPWR_0_IO_UNUSED2
OPA4197IPWR
U3403C
CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
CORE_PACK_OPA4197IPWR_0_IO_UNUSED3
OPA4197IPWR
U3403D
CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
CORE_PACK_OPA4197IPWR_0_IO_UNUSED4
OPA4197IPWR
U3403E
V+
V-
OPA4197IPWR
C3402
100 nF
C3403
100 nF
C3404
100 nF
C3405
100 nF
C3406
100 nF
C3407
100 nF
R3417
C025c3f_SENSE
4990 Ω
D3403
C025c3f_DRIVE
C025c3f_LED_A
BAS16GW-QX
D3404
C025c3f_SENSE
C025c3f_LED_A
BAS16GW-QX
D3405
C025c3f_LED_K
C025c3f_DRIVE
BAS16GW-QX
D3406
C025c3f_LED_K
C025c3f_SENSE
BAS16GW-QX
C025c3f_LED_A
C025c3f_LED_K
KT-0603W
R3418
Cc6c62b_SENSE
4990 Ω
D3408
Cc6c62b_DRIVE
Cc6c62b_LED_A
BAS16GW-QX
D3409
Cc6c62b_SENSE
Cc6c62b_LED_A
BAS16GW-QX
D3410
Cc6c62b_LED_K
Cc6c62b_DRIVE
BAS16GW-QX
D3411
Cc6c62b_LED_K
Cc6c62b_SENSE
BAS16GW-QX
Cc6c62b_LED_A
Cc6c62b_LED_K
KT-0603W
R3419
C7971de_SENSE
4990 Ω
D3413
C7971de_DRIVE
C7971de_LED_A
BAS16GW-QX
D3414
C7971de_SENSE
C7971de_LED_A
BAS16GW-QX
D3415
C7971de_LED_K
C7971de_DRIVE
BAS16GW-QX
D3416
C7971de_LED_K
C7971de_SENSE
BAS16GW-QX
D1010
C7971de_LED_A
C7971de_LED_K
KT-0603W
U3404A
A_BUFFER
C025c3f_SENSE
C025c3f_DRIVE
OPA4196IDR
U3404B
B_BUFFER
Cc6c62b_SENSE
Cc6c62b_DRIVE
OPA4196IDR
U3404C
OUT_BUFFERED
C7971de_SENSE
C7971de_DRIVE
OPA4196IDR
U3404D
LED_PACK_OPA4196IDR_0_IO_UNUSED4
LED_PACK_OPA4196IDR_0_IO_UNUSED4
OPA4196IDR
U3404E
OPA4196IDR
C3408
100 nF
C3409
100 nF
U3405A
A_SENSE
A_BUFFER
A_BUFFER
OPA4197IPWR
U3405B
B_SENSE
B_BUFFER
B_BUFFER
OPA4197IPWR
U3405C
SELECTOR_REMOTE
C61b194_FB
OUT_BUFFERED
OPA4197IPWR
U3405D
IO_EXTRA_5_IO_UNUSED4
IO_EXTRA_5_IO_UNUSED4
OPA4197IPWR
U3405E
OPA4197IPWR
C3410
100 nF
C3411
100 nF

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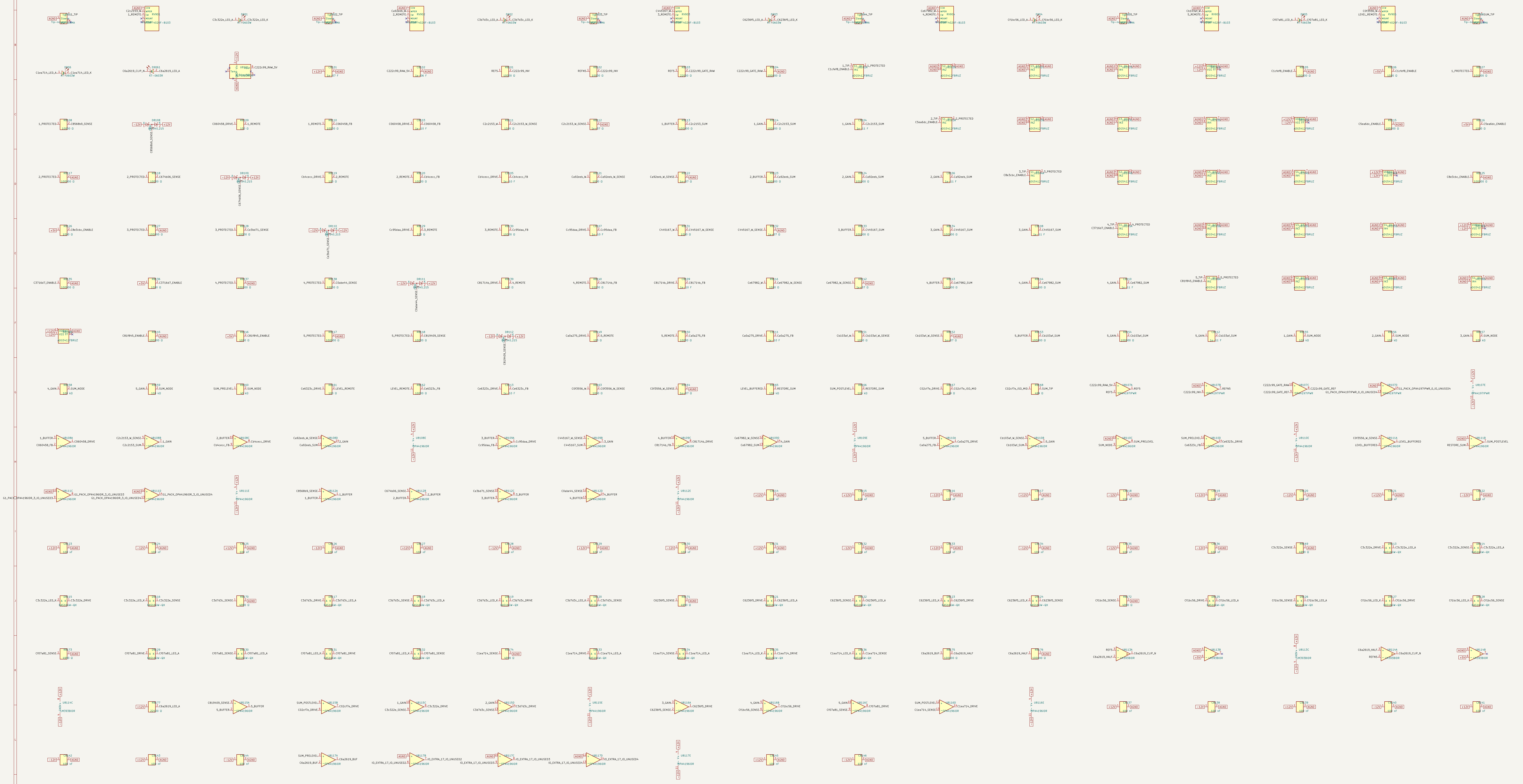

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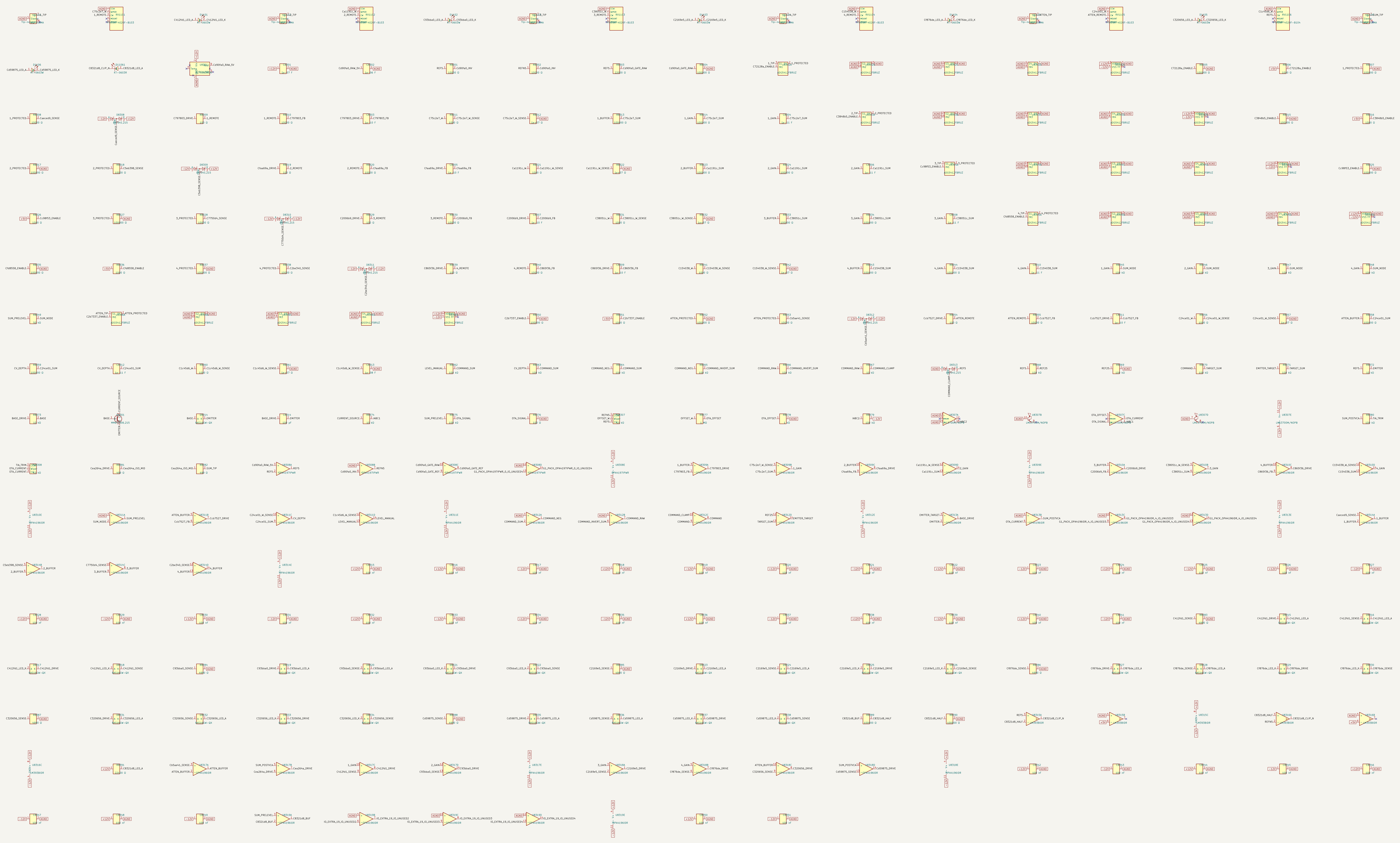

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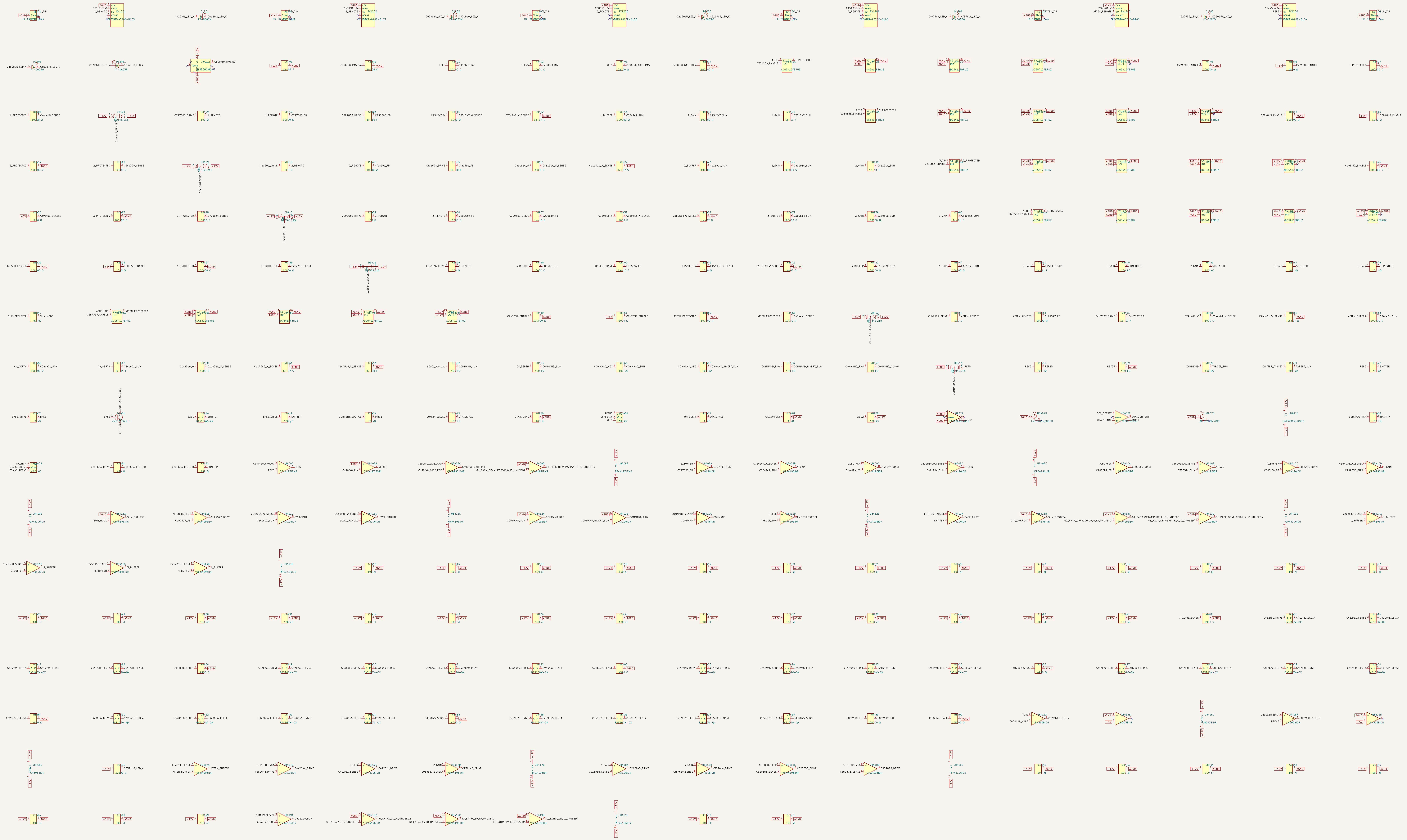

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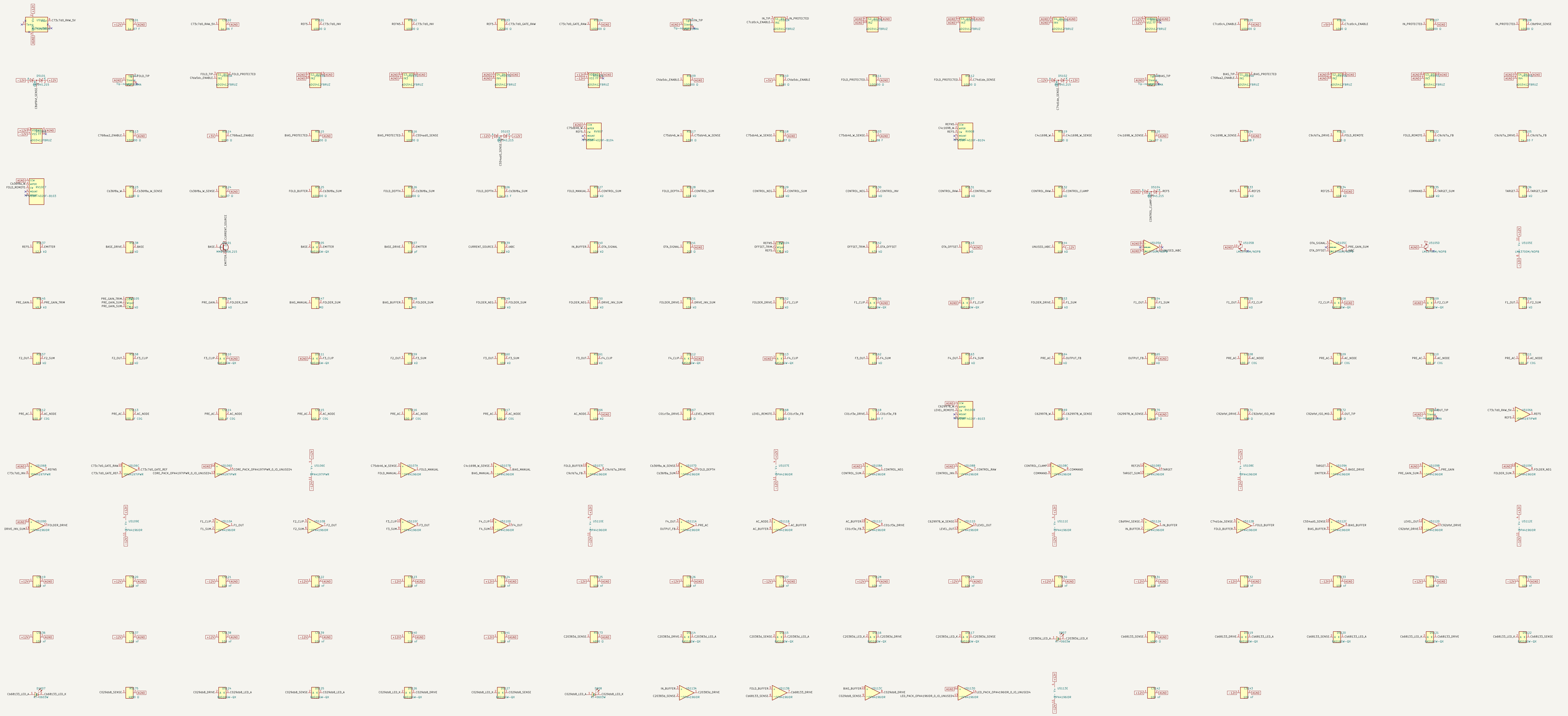

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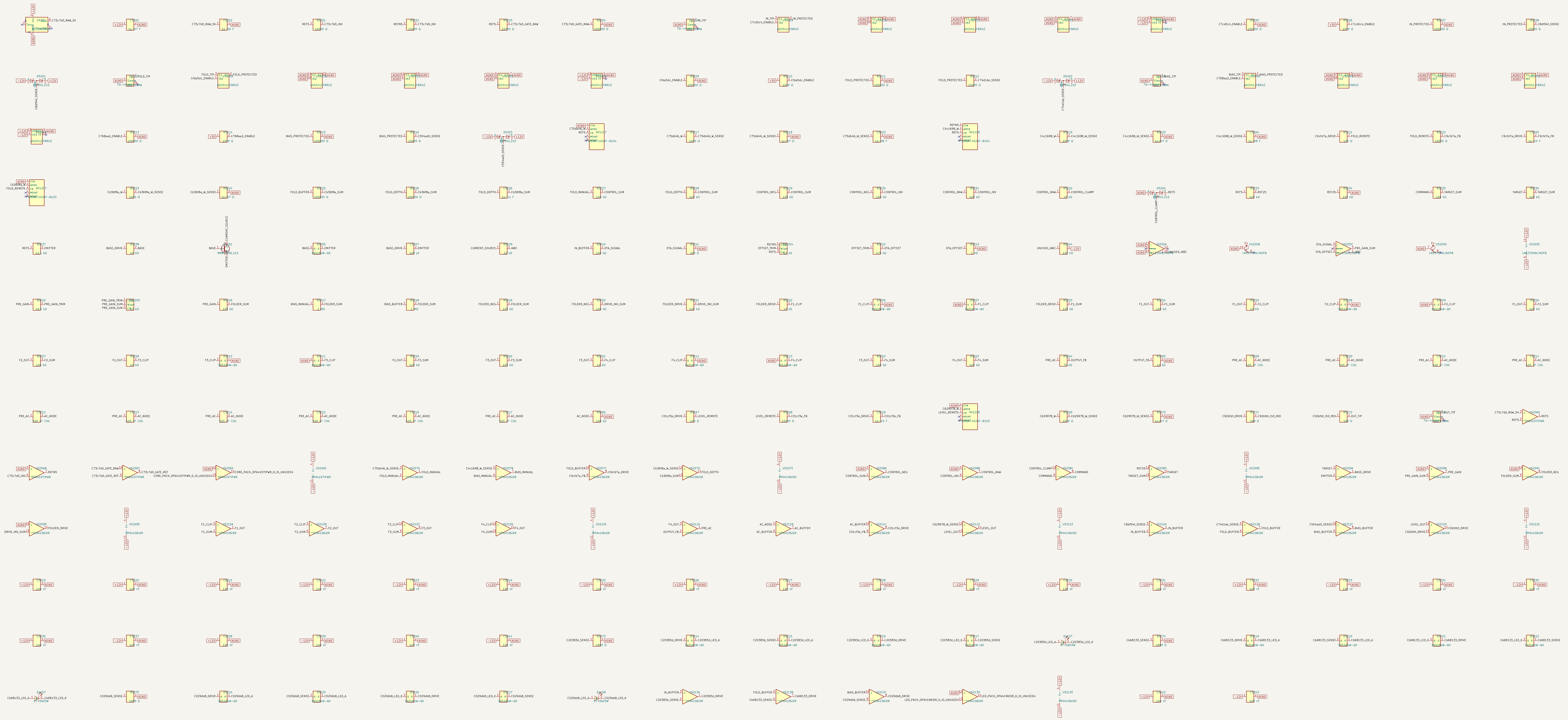

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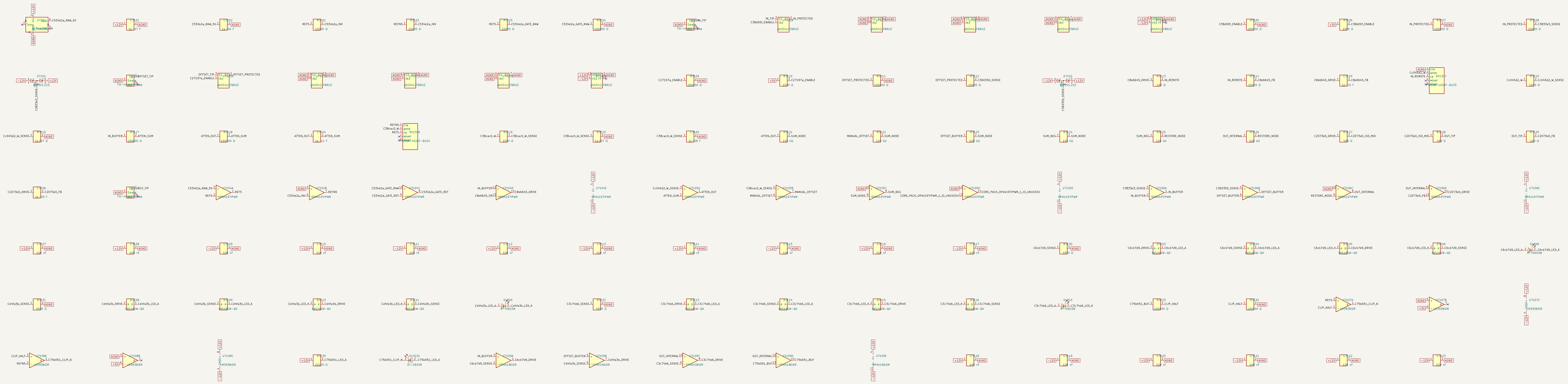

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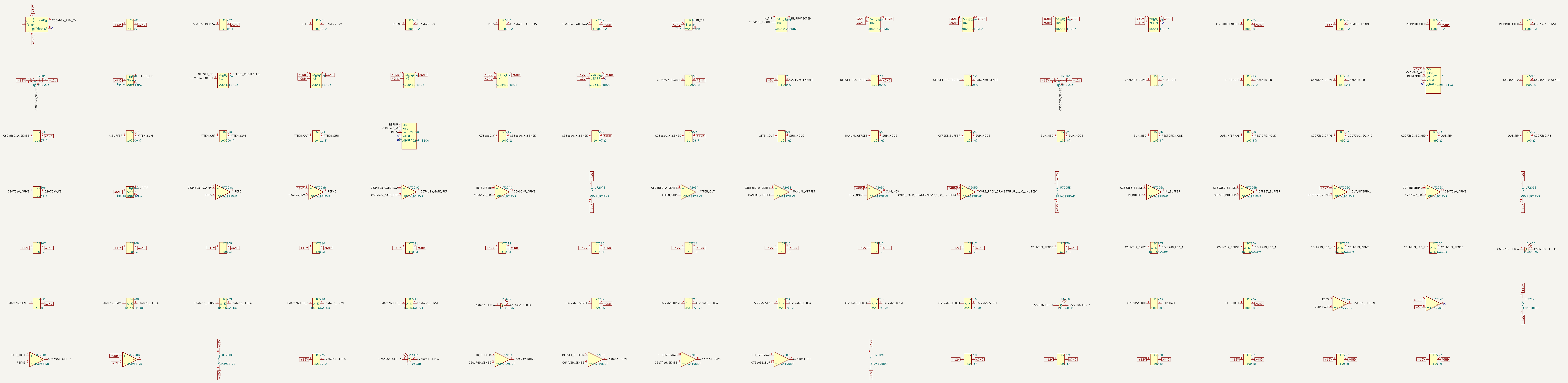
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C7201 +12V AGND 1e-07 F
C7202 C534b2a_RAW_5V AGND 1e-06 F
R7201 REF5 C534b2a_INV 10000 Ω
R7202 REFN5 C534b2a_INV 10000 Ω
R7203 REF5 C534b2a_GATE_RAW 22000 Ω
R7204 C534b2a_GATE_RAW AGND 100000 Ω
Tip Sleeve TN AGND IN_TIP Tip-normal
U7202A ADG5412FBRUZ S1 D1 IN1 IN_TIP IN_PROTECTED C38d00f_ENABLE
U7202B ADG5412FBRUZ S2 D2 IN2 AGND
U7202C ADG5412FBRUZ S3 D3 IN3 AGND
U7202D ADG5412FBRUZ S4 D4 IN4 AGND
R7205 C38d00f_ENABLE AGND 100000 Ω
R7206 +5V C38d00f_ENABLE 1000 Ω
R7207 IN_PROTECTED AGND 100000 Ω
R7208 IN_PROTECTED C3833e3_SENSE 10000 Ω
D7201 BAT54S,215 −12V +12V A K COM C3833e3_SENSE
Tip Sleeve TN AGND OFFSET_TIP Tip-normal
U7203A ADG5412FBRUZ S1 D1 IN1 OFFSET_TIP OFFSET_PROTECTED C27197a_ENABLE
U7203B ADG5412FBRUZ S2 D2 IN2 AGND
U7203C ADG5412FBRUZ S3 D3 IN3 AGND
U7203D ADG5412FBRUZ S4 D4 IN4 AGND
U7203E ADG5412FBRUZ VDD VSS GND +12V −12V AGND
R7209 C27197a_ENABLE AGND 100000 Ω
R7210 +5V C27197a_ENABLE 1000 Ω
R7211 OFFSET_PROTECTED AGND 100000 Ω
R7212 OFFSET_PROTECTED C360350_SENSE 10000 Ω
D7202 BAT54S,215 −12V +12V A K COM C360350_SENSE
R7213 C8e6645_DRIVE IN_REMOTE 100 Ω
R7214 IN_REMOTE C8e6645_FB 10000 Ω
C7203 C8e6645_DRIVE C8e6645_FB 1e-10 F
RV1407 PTV09A-4020F-B103 CCW WIPER CW MOUNT AGND Cc045d2_W IN_REMOTE
R7215 Cc045d2_W Cc045d2_W_SENSE 1000 Ω
R7216 Cc045d2_W_SENSE AGND 1e+07 Ω
R7217 IN_BUFFER ATTEN_SUM 100000 Ω
R7218 ATTEN_OUT ATTEN_SUM 100000 Ω
C7204 ATTEN_OUT ATTEN_SUM 1e-11 F
RV1408 PTV09A-4020F-B104 CCW WIPER CW MOUNT REFN5 C38cac0_W REF5
R7219 C38cac0_W C38cac0_W_SENSE 1000 Ω
R7220 C38cac0_W_SENSE AGND 1e+07 Ω
C7205 C38cac0_W_SENSE AGND 1e-08 F
R7221 ATTEN_OUT SUM_NODE 100 kΩ
R7222 MANUAL_OFFSET SUM_NODE 100 kΩ
R7223 OFFSET_BUFFER SUM_NODE 100 kΩ
R7224 SUM_NEG SUM_NODE 100 kΩ
R7225 SUM_NEG RESTORE_NODE 100 kΩ
R7226 OUT_INTERNAL RESTORE_NODE 100 kΩ
R7227 C2073e0_DRIVE C2073e0_ISO_MID 499 Ω
R7228 C2073e0_ISO_MID OUT_TIP 499 Ω
R7229 OUT_TIP C2073e0_FB 100 Ω
C7206 C2073e0_DRIVE C2073e0_FB 1e-09 F
Tip Sleeve TN AGND OUT_TIP Tip-normal
U7204A OPA4197IPWR C534b2a_RAW_5V REF5 REF5
U7204B OPA4197IPWR AGND C534b2a_INV REFN5
U7204C OPA4197IPWR C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF
U7204D OPA4197IPWR IN_BUFFER C8e6645_FB C8e6645_DRIVE
U7204E OPA4197IPWR V+ V− +12V −12V
U7205A OPA4197IPWR Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT
U7205B OPA4197IPWR C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET
U7205C OPA4197IPWR AGND SUM_NODE SUM_NEG
U7205D OPA4197IPWR AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4
U7205E OPA4197IPWR V+ V− +12V −12V
U7206A OPA4197IPWR C3833e3_SENSE IN_BUFFER IN_BUFFER
U7206B OPA4197IPWR C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER
U7206C OPA4197IPWR AGND RESTORE_NODE OUT_INTERNAL
U7206D OPA4197IPWR OUT_INTERNAL C2073e0_FB C2073e0_DRIVE
U7206E OPA4197IPWR V+ V− +12V −12V
C7207 +12V AGND 100 nF
C7208 +12V AGND 100 nF
C7209 −12V AGND 100 nF
C7210 +12V AGND 100 nF
C7211 −12V AGND 100 nF
C7212 +12V AGND 100 nF
C7213 −12V AGND 100 nF
C7214 +12V AGND 100 nF
C7215 −12V AGND 100 nF
C7216 +12V AGND 100 nF
C7217 −12V AGND 100 nF
R7230 C6cb7d9_SENSE AGND 4990 Ω
D7203 BAS16GW-QX A K C6cb7d9_DRIVE C6cb7d9_LED_A
D7204 BAS16GW-QX A K C6cb7d9_SENSE C6cb7d9_LED_A
D7205 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_DRIVE
D7206 BAS16GW-QX A K C6cb7d9_LED_K C6cb7d9_SENSE
D1408 KT-0603W C6cb7d9_LED_A C6cb7d9_LED_K
R7231 Cd4fa3b_SENSE AGND 4990 Ω
D7208 BAS16GW-QX A K Cd4fa3b_DRIVE Cd4fa3b_LED_A
D7209 BAS16GW-QX A K Cd4fa3b_SENSE Cd4fa3b_LED_A
D7210 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_DRIVE
D7211 BAS16GW-QX A K Cd4fa3b_LED_K Cd4fa3b_SENSE
D1409 KT-0603W Cd4fa3b_LED_A Cd4fa3b_LED_K
R7232 C3c74b6_SENSE AGND 4990 Ω
D7213 BAS16GW-QX A K C3c74b6_DRIVE C3c74b6_LED_A
D7214 BAS16GW-QX A K C3c74b6_SENSE C3c74b6_LED_A
D7215 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_DRIVE
D7216 BAS16GW-QX A K C3c74b6_LED_K C3c74b6_SENSE
D1410 KT-0603W C3c74b6_LED_A C3c74b6_LED_K
R7233 C75b051_BUF CLIP_HALF 100000 Ω
R7234 CLIP_HALF AGND 100000 Ω
U7207A LM393BIDR REF5 CLIP_HALF C75b051_CLIP_N
U7207B LM393BIDR AGND +5V
U7207C LM393BIDR V+ GND +12V −12V
U7208A LM393BIDR CLIP_HALF REFN5 C75b051_CLIP_N
U7208B LM393BIDR AGND +5V
U7208C LM393BIDR V+ GND +12V −12V
R7235 +12V C75b051_LED_A 22100 Ω
D14101 KT-0603R C75b051_CLIP_N C75b051_LED_A
U7209A OPA4196IDR IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE
U7209B OPA4196IDR OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE
U7209C OPA4196IDR OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE
U7209D OPA4196IDR OUT_INTERNAL C75b051_BUF C75b051_BUF
U7209E OPA4196IDR V+ V− +12V −12V
C7218 +12V AGND 100 nF
C7219 −12V AGND 100 nF
C7220 +12V AGND 100 nF
C7221 −12V AGND 100 nF
C7222 +12V AGND 100 nF
C7223 −12V AGND 100 nF

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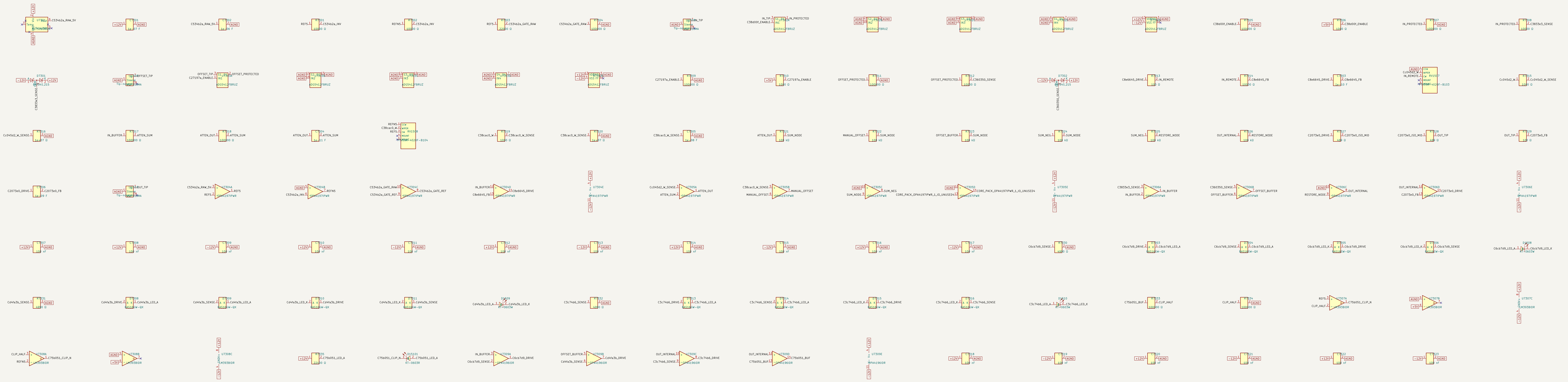
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C7301 +12V 1e-07 F AGND
C7302 C534b2a_RAW_5V 1e-06 F AGND
R7301 REF5 10000 Ω C534b2a_INV
R7302 REFN5 10000 Ω C534b2a_INV
R7303 REF5 22000 Ω C534b2a_GATE_RAW
R7304 C534b2a_GATE_RAW 100000 Ω AGND
AGND Tip Sleeve Tip-normal IN_TIP
U7302A IN_TIP C38d00f_ENABLE S1 D1 IN1 IN_PROTECTED ADG5412FBRUZ
U7302B AGND AGND S2 D2 IN2 AGND ADG5412FBRUZ
U7302C AGND AGND S3 D3 IN3 AGND ADG5412FBRUZ
U7302D AGND AGND S4 D4 IN4 AGND ADG5412FBRUZ
U7302E +12V -12V VDD GND VSS FF AGND ADG5412FBRUZ
R7305 C38d00f_ENABLE 100000 Ω AGND
R7306 +5V 1000 Ω C38d00f_ENABLE
R7307 IN_PROTECTED 100000 Ω AGND
R7308 IN_PROTECTED 10000 Ω C3833e3_SENSE
D7301 -12V +12V BAT54S,215 C3833e3_SENSE
AGND Tip Sleeve Tip-normal OFFSET_TIP
U7303A OFFSET_TIP C27197a_ENABLE S1 D1 IN1 OFFSET_PROTECTED ADG5412FBRUZ
U7303B AGND AGND S2 D2 IN2 AGND ADG5412FBRUZ
U7303C AGND AGND S3 D3 IN3 AGND ADG5412FBRUZ
U7303D AGND AGND S4 D4 IN4 AGND ADG5412FBRUZ
U7303E +12V -12V VDD GND VSS FF AGND ADG5412FBRUZ
R7309 C27197a_ENABLE 100000 Ω AGND
R7310 +5V 1000 Ω C27197a_ENABLE
R7311 OFFSET_PROTECTED 100000 Ω AGND
R7312 OFFSET_PROTECTED 10000 Ω C360350_SENSE
D7302 -12V +12V BAT54S,215 C360350_SENSE
R7313 C8e6645_DRIVE 100 Ω IN_REMOTE
R7314 IN_REMOTE 10000 Ω C8e6645_FB
C7303 C8e6645_DRIVE 1e-10 F C8e6645_FB
RV1507 AGND Cc045d2_W IN_REMOTE CCW WIPER CW MOUNT MOUNT 4020F-B103
R7315 Cc045d2_W 1000 Ω Cc045d2_W_SENSE
R7316 Cc045d2_W_SENSE 1e+07 Ω AGND
R7317 IN_BUFFER 100000 Ω ATTEN_SUM
R7318 ATTEN_OUT 100000 Ω ATTEN_SUM
C7304 ATTEN_OUT 1e-11 F ATTEN_SUM
RV1508 REFN5 C38cac0_W REF5 CCW WIPER CW MOUNT MOUNT 4020F-B104
R7319 C38cac0_W 1000 Ω C38cac0_W_SENSE
R7320 C38cac0_W_SENSE 1e+07 Ω AGND
C7305 C38cac0_W_SENSE 1e-08 F AGND
R7321 ATTEN_OUT 100 kΩ SUM_NODE
R7322 MANUAL_OFFSET 100 kΩ SUM_NODE
R7323 OFFSET_BUFFER 100 kΩ SUM_NODE
R7324 SUM_NEG 100 kΩ SUM_NODE
R7325 SUM_NEG 100 kΩ RESTORE_NODE
R7326 OUT_INTERNAL 100 kΩ RESTORE_NODE
R7327 C2073e0_DRIVE 499 Ω C2073e0_ISO_MID
R7328 C2073e0_ISO_MID 499 Ω OUT_TIP
R7329 OUT_TIP 100 Ω C2073e0_FB
C7306 C2073e0_DRIVE 1e-09 F C2073e0_FB
AGND Tip Sleeve Tip-normal OUT_TIP
U7304A C534b2a_RAW_5V REF5 REF5 OPA4197IPWR
U7304B AGND C534b2a_INV REFN5 OPA4197IPWR
U7304C C534b2a_GATE_RAW C534b2a_GATE_REF C534b2a_GATE_REF OPA4197IPWR
U7304D IN_BUFFER C8e6645_FB C8e6645_DRIVE OPA4197IPWR
U7304E +12V -12V V+ V- OPA4197IPWR
U7305A Cc045d2_W_SENSE ATTEN_SUM ATTEN_OUT OPA4197IPWR
U7305B C38cac0_W_SENSE MANUAL_OFFSET MANUAL_OFFSET OPA4197IPWR
U7305C AGND SUM_NODE SUM_NEG OPA4197IPWR
U7305D AGND CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 CORE_PACK_OPA4197IPWR_1_IO_UNUSED4 OPA4197IPWR
U7305E +12V -12V V+ V- OPA4197IPWR
U7306A C3833e3_SENSE IN_BUFFER IN_BUFFER OPA4197IPWR
U7306B C360350_SENSE OFFSET_BUFFER OFFSET_BUFFER OPA4197IPWR
U7306C AGND RESTORE_NODE OUT_INTERNAL OPA4197IPWR
U7306D OUT_INTERNAL C2073e0_FB C2073e0_DRIVE OPA4197IPWR
U7306E +12V -12V V+ V- OPA4197IPWR
C7307 +12V 100 nF AGND
C7308 +12V 100 nF AGND
C7309 -12V 100 nF AGND
C7310 +12V 100 nF AGND
C7311 -12V 100 nF AGND
C7312 +12V 100 nF AGND
C7313 -12V 100 nF AGND
C7314 +12V 100 nF AGND
C7315 -12V 100 nF AGND
C7316 +12V 100 nF AGND
C7317 -12V 100 nF AGND
R7330 C6cb7d9_SENSE 4990 Ω AGND
D7303 C6cb7d9_DRIVE A K C6cb7d9_LED_A BAS16GW-QX
D7304 C6cb7d9_SENSE A K C6cb7d9_LED_A BAS16GW-QX
D7305 C6cb7d9_LED_K A K C6cb7d9_DRIVE BAS16GW-QX
D7306 C6cb7d9_LED_K A K C6cb7d9_SENSE BAS16GW-QX
D1608 C6cb7d9_LED_A C6cb7d9_LED_K KT-0603W
R7331 Cd4fa3b_SENSE 4990 Ω AGND
D7308 Cd4fa3b_DRIVE A K Cd4fa3b_LED_A BAS16GW-QX
D7309 Cd4fa3b_SENSE A K Cd4fa3b_LED_A BAS16GW-QX
D7310 Cd4fa3b_LED_K A K Cd4fa3b_DRIVE BAS16GW-QX
D7311 Cd4fa3b_LED_K A K Cd4fa3b_SENSE BAS16GW-QX
D1609 Cd4fa3b_LED_A Cd4fa3b_LED_K KT-0603W
R7332 C3c74b6_SENSE 4990 Ω AGND
D7313 C3c74b6_DRIVE A K C3c74b6_LED_A BAS16GW-QX
D7314 C3c74b6_SENSE A K C3c74b6_LED_A BAS16GW-QX
D7315 C3c74b6_LED_K A K C3c74b6_DRIVE BAS16GW-QX
D7316 C3c74b6_LED_K A K C3c74b6_SENSE BAS16GW-QX
D1610 C3c74b6_LED_A C3c74b6_LED_K KT-0603W
R7333 C75b051_BUF 100000 Ω CLIP_HALF
R7334 CLIP_HALF 100000 Ω AGND
U7307A REF5 CLIP_HALF C75b051_CLIP_N LM393BIDR
U7307B AGND +5V LM393BIDR
U7307C +12V -12V GND V+ LM393BIDR
U7308A CLIP_HALF REFN5 C75b051_CLIP_N LM393BIDR
U7308B AGND +5V LM393BIDR
U7308C +12V -12V GND V+ LM393BIDR
R7335 +12V 22100 Ω C75b051_LED_A
D15101 C75b051_CLIP_N C75b051_LED_A KT-0603R
U7309A IN_BUFFER C6cb7d9_SENSE C6cb7d9_DRIVE OPA4196IDR
U7309B OFFSET_BUFFER Cd4fa3b_SENSE Cd4fa3b_DRIVE OPA4196IDR
U7309C OUT_INTERNAL C3c74b6_SENSE C3c74b6_DRIVE OPA4196IDR
U7309D OUT_INTERNAL C75b051_BUF C75b051_BUF OPA4196IDR
U7309E +12V -12V V+ V- OPA4196IDR
C7318 +12V 100 nF AGND
C7319 -12V 100 nF AGND
C7320 +12V 100 nF AGND
C7321 -12V 100 nF AGND
C7322 +12V 100 nF AGND
C7323 -12V 100 nF AGND

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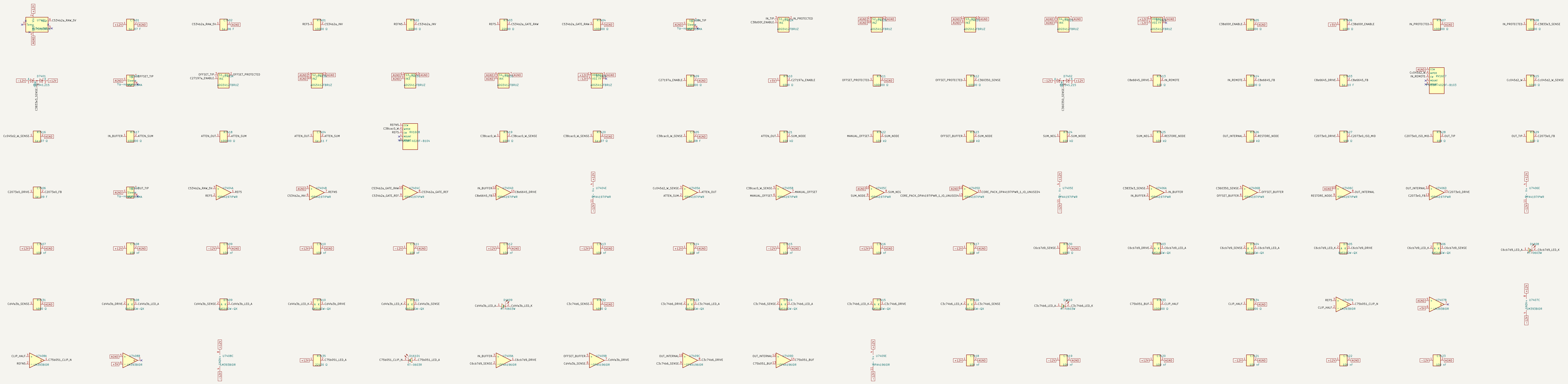

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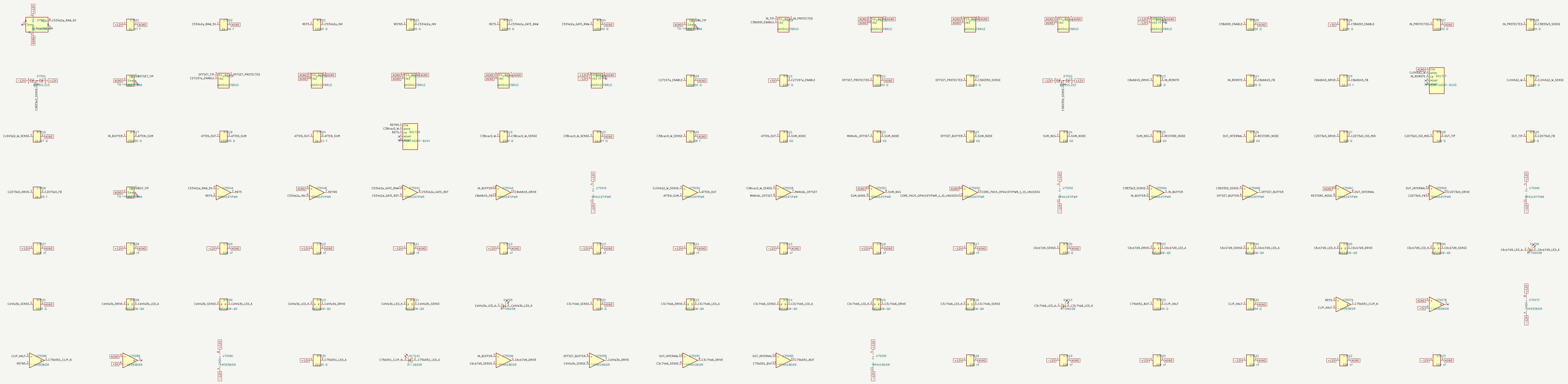

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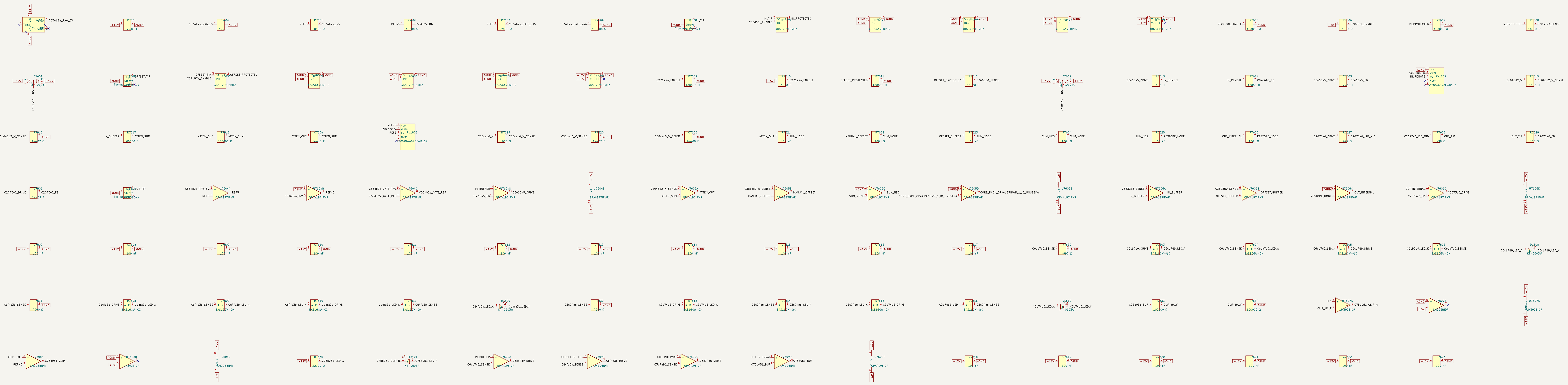

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